

Geometric State Fusion for Autonomous Agents: A Comparative Analysis of Dual Quaternion Observer and Kalman Filters

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MOTIVATION

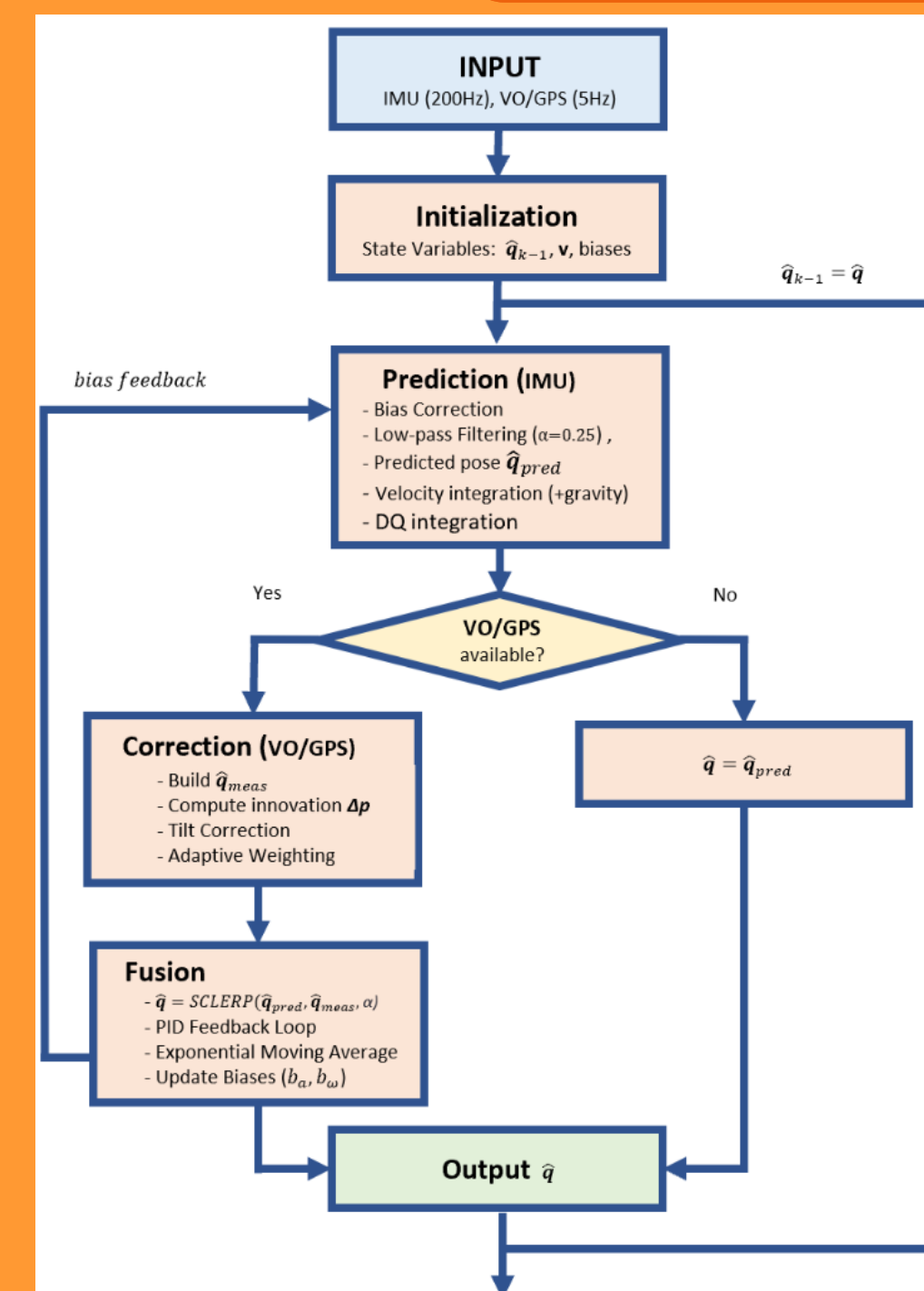
Our motivation

- ❑ **The Challenge:** Reliable 6DoF pose estimation is critical for IoT edge devices and autonomous robot systems.
- ❑ **Limitations of Filter-Based Methods:**
 - Traditional (EKF / ESKF): Artificial decoupling of rotation and translation, leading to linearization artifacts and drift.
 - Manifold-Aware (UKF-M): Complex sigma-point transformations make it computationally prohibitive for low-power edge hardware.
- ❑ **The Geometric Gap:** Standard frameworks often trade geometric fidelity for mathematical convenience.

Abstract

Autonomous agents operating in dynamic 3D environments require robust pose estimation that preserves the geometric structure of rigid-body motion. Traditional filtering approaches often decouple rotation and translation, leading to kinematic inconsistencies. This paper presents a comprehensive benchmark of three geometrically-aware architectures: a **Dual Quaternion Geometric Observer (GeoDQ)**, a manifold-aware Unscented Kalman Filter (UKF-M), and an Error-State Kalman Filter (ESKF). We evaluate these on the complete RoNIN dataset (35 trajectories). Results demonstrate that the proposed GeoDQ method significantly outperforms filtering baselines, reducing RMSE by 3.7× compared to ESKF and 6.1× compared to UKF-M. Furthermore, robustness analysis reveals that the geometric observer maintains superior stability under sparse updates. Despite its mathematical rigor, GeoDQ is the most computationally efficient, executing 10% faster than ESKF and 5× faster than UKF-M, making it ideal for resource-constrained embedded agents.

METHODOLOGY



GeoDQ Observer Architecture

Algorithm 1 GeoDQ Geometric Observer

```

1: Input:  $\omega, a$  (IMU),  $p_{\text{vo}}, q_{\text{vo}}$  (VO), Params:  $dt, \alpha_{\text{base}}, k_{ba}$ 
2: for each IMU sample do
3:    $\hat{q} \leftarrow \hat{q} \odot dq_{\text{exp}}((\omega - b_g)dt, v_b dt)$ 
4:    $v \leftarrow v + (R(\hat{q})(a - b_a) + g)dt$ 
5:   if VO available then
6:      $\Delta p \leftarrow p_{\text{vo}} - p(\hat{q})$ 
7:      $\hat{q} \leftarrow \text{SCLERP}(\hat{q}, q_{\text{vo}}, \alpha)$ 
8:      $b_a \leftarrow b_a - k_{ba} \cdot (R(\hat{q})^T \Delta p)$ 
9:   end if
10: end for

```

- ❑ **Dual Quaternion Representation:** $\hat{q} = q_r + \varepsilon \cdot q_d$
- ❑ IMU prediction is corrected via SCLERP fusion with bias feedback loops
- ❑ **SCLERP Fusion Mechanism:**
 $\hat{q}_{\text{new}} = \hat{q}_{\text{pred}} \otimes (\hat{q}_{\text{pred}} \otimes \hat{q}_{\text{vo}})^{\alpha_{\text{rob}}}$
- ❑ **Key Features:**
 - Coordinate-free interpolation without Jacobians.
 - Adaptive Huber-weighted innovation gating (α_{rob}).
 - PID-based continuous bias feedback loop.

EXPERIMENTS

Large-Scale Evaluation on RoNIN Dataset: Observer (GeoDQ) vs. Filters

Trajectory Analysis

Sequence: a000_11

Method	RMSE[m]	ATE [m]	Time [ms]
ESKF (JIT)	0.1548	0.0697	2019.1
UKF-M (JIT)	0.2642	0.1154	9117.2
GeoDQ (ours)	0.0051	0.0046	1787.2

Overall RoNIN Benchmark (35 Trajectories)

GeoDQ RMSE: 0.0043 ± 0.0009 m

GeoDQ Speed: 3.05 s per traj.

Large-Scale Evaluation on RoNIN Dataset: Observer (GeoDQ) vs. Filters

Trajectory Analysis

Sequence: a025_2

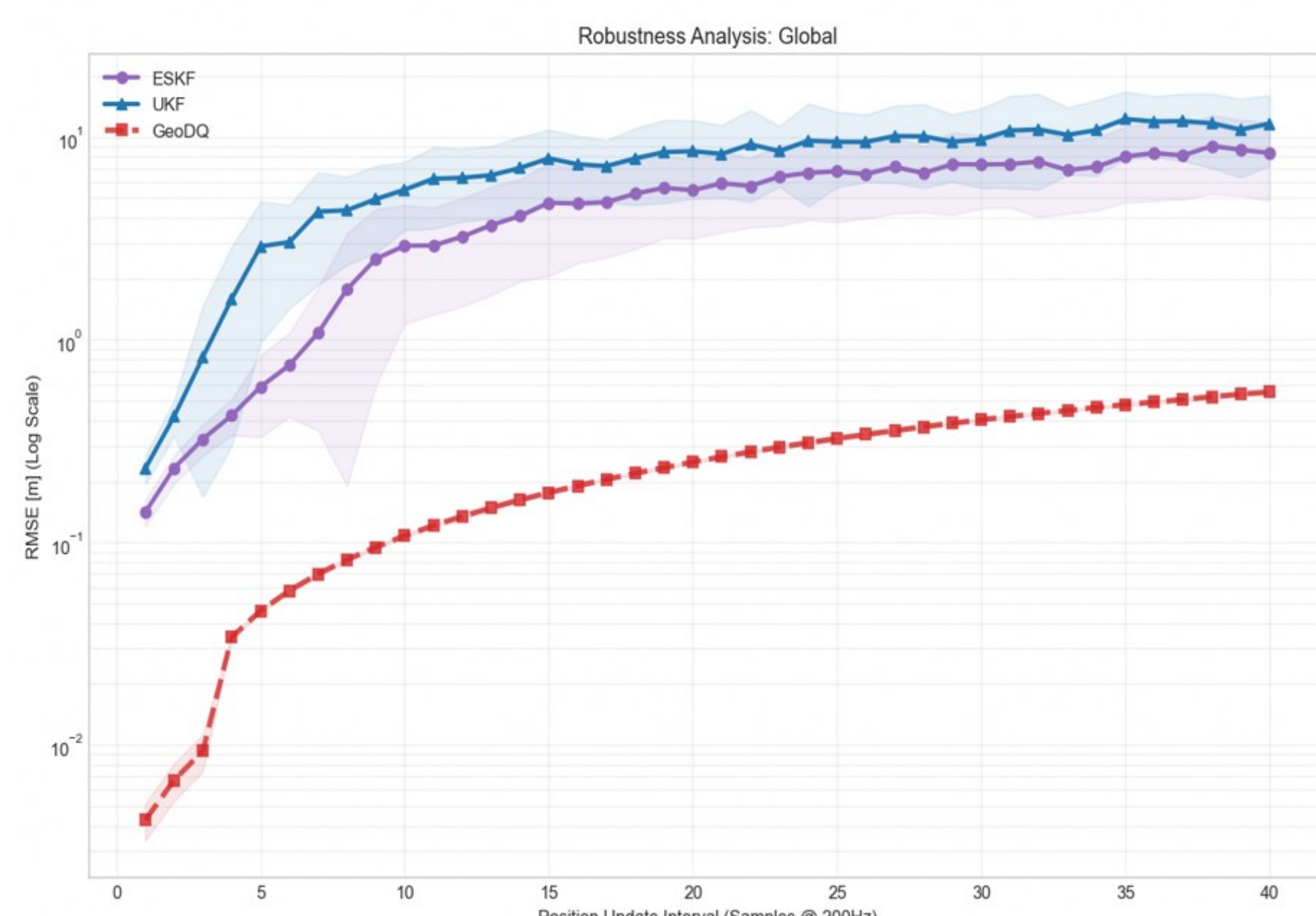
Method	RMSE[m]	ATE [m]	Time [ms]
ESKF (JIT)	0.1461	0.1016	3455.2
UKF-M (JIT)	0.2118	0.1358	15160.9
GeoDQ (ours)	0.0052	0.0050	3055.9

Overall RoNIN Benchmark (35 Trajectories)

GeoDQ RMSE: 0.0043 ± 0.0009 m

GeoDQ Speed: 3.05 s per traj.

Global Robustness Analysis



Resilience to Sensor Dropouts:

- Simulated Visual Odometry update gaps (up to 40x decimation).
- Filters (ESKF/UKF-M): Rapid covariance inflation and divergence.
- Observer (GeoDQ): Smooth, logarithmic error degradation. SCLERP acts as a stabilizing "geometric spring".

Performance Benchmark (35 RoNIN Trajectories)

Method	RMSE [m] (Mean ± Std)	Execution [ms]
ESKF (Standard)	0.1409 ± 0.0213	11469.7
ESKF (JIT)	0.1409 ± 0.0213	3463.7
UKF-M (Standard)	0.2329 ± 0.0382	45753.0
UKF-M (JIT)	0.2312 ± 0.0378	15211.0
GeoDQ (Standard)	0.0043 ± 0.0009	23472.6
GeoDQ (JIT)	0.0043 ± 0.0009	3048.3

Key Takeaways:

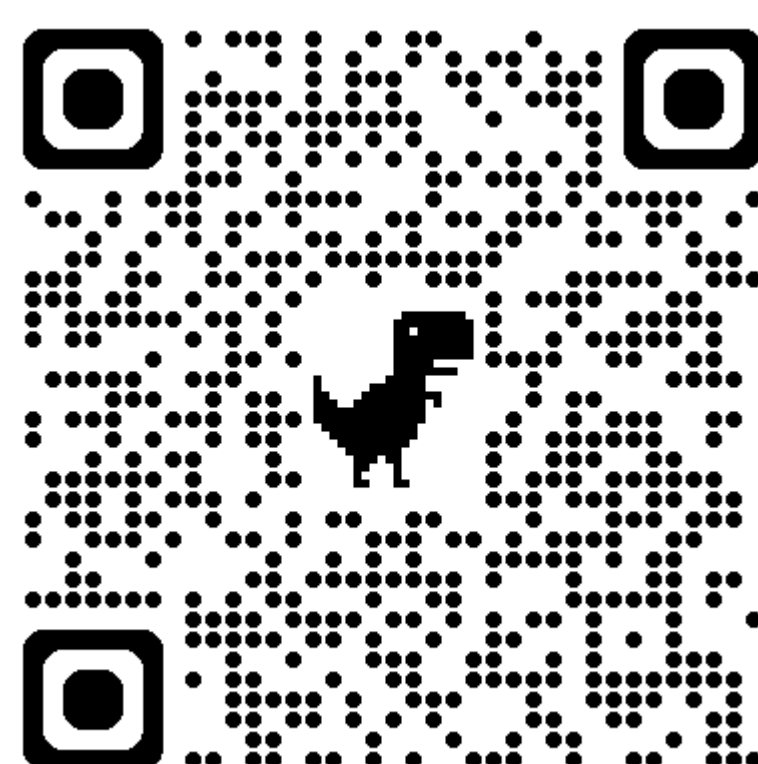
- ❑ **Superior Accuracy:** Lowest RMSE (0.0043 m) with minimal standard deviation.
- ❑ **JIT Acceleration:** ~7× speedup over standard Python.
- ❑ **Efficiency:** Outperforms filter-based methods by orders of magnitude.

CONCLUSION

Summary

GeoDQ is a dual quaternion geometric observer that unifies rotation and translation on the SE(3) manifold, using SCLERP interpolation for IMU-VO fusion without linearization artifacts, making it suitable for resource-constrained agents.

- ❑ **Open-Source Implementation** available at: <https://afanasyev.github.io/SE3-Manifold-Lib/>



Key Contributions

- ❑ **Geometric SE(3) Formulation:** Elimination of linearization drift via dual quaternions.
- ❑ **Adaptive Fusion:** Robust SCLERP mechanism with heuristic tilt correction.
- ❑ **High Efficiency:** JIT-compiled architecture suitable for edge devices.
- ❑ **Proven Robustness:** Unmatched stability during unconstrained motion and sensor dropouts.

Implications for Multi-Agent Systems

- GeoDQ's low computational footprint enables parallel independent state estimation across multiple agents, while SE(3) geometric consistency simplifies relative localization and formation control. Its robustness to sensor reorientation and reduced reliance on frequent centralized updates make it well-suited for distributed multi-robot systems operating in GPS-denied environments.

Limitations – Dependency on Signal Quality

- Relies on geometric interpolation; assumes robust VO signal.
- Lacks probabilistic covariance to gate extremely noisy outliers.

Future Work

- Integrating lightweight probabilistic metrics into the interpolation factor α .
- Empirical validation on high-speed platforms (e.g., agile drones) under extreme accelerations.